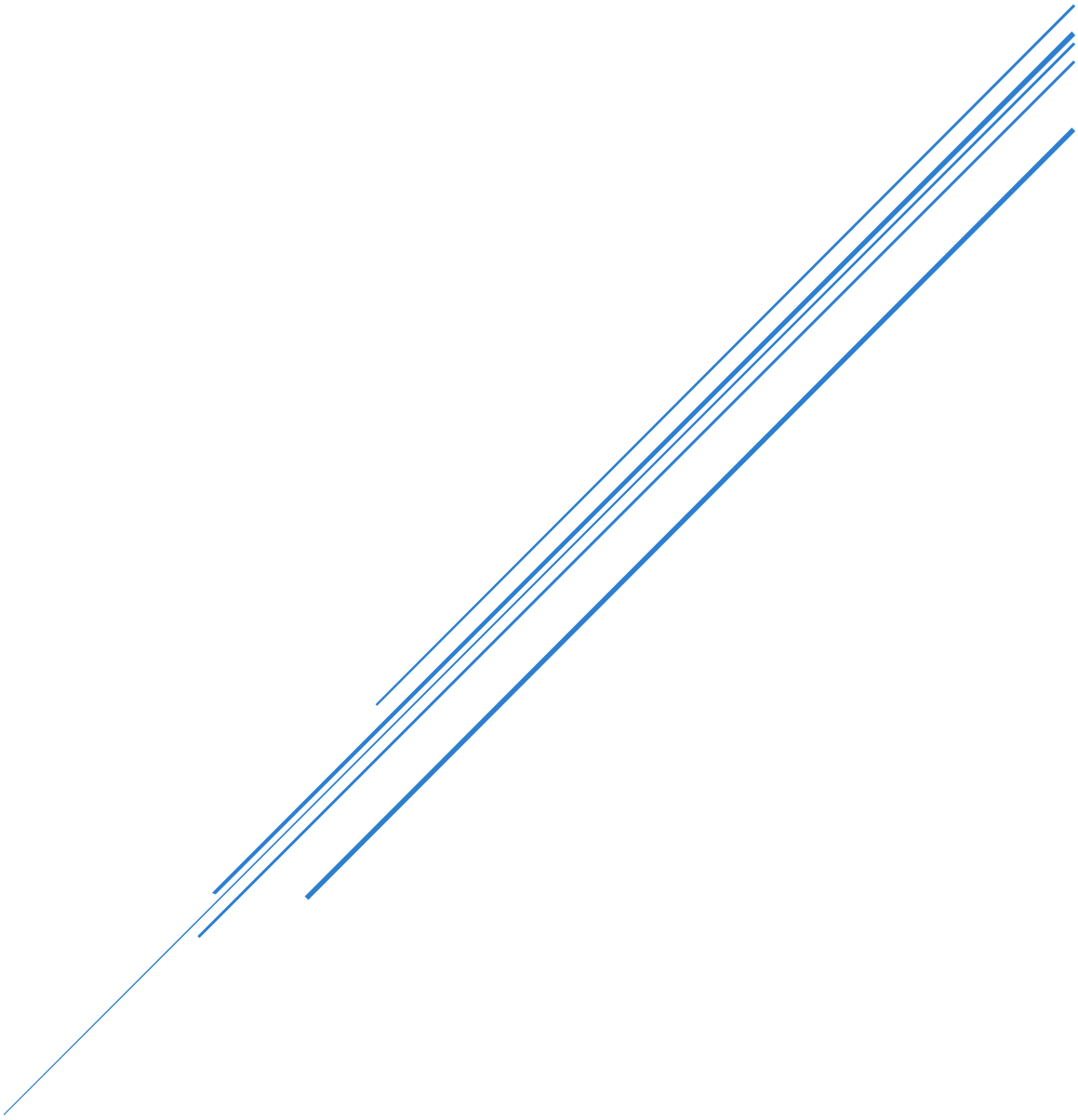
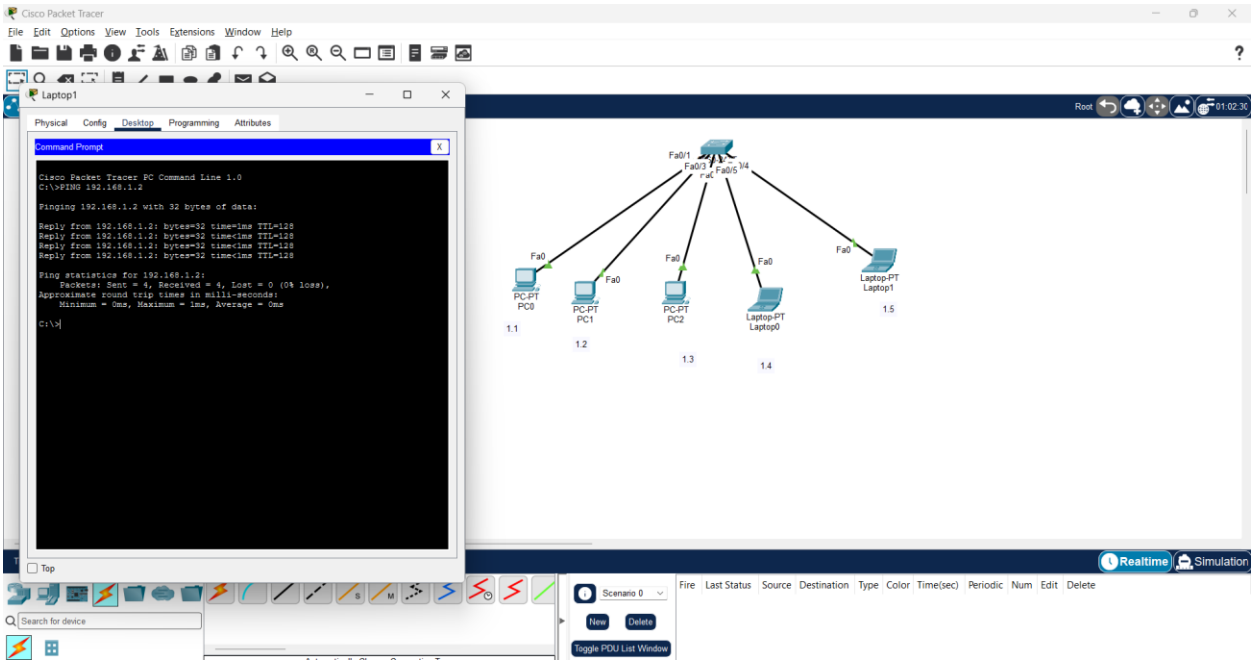
DAY - 1- 5 BUILD AND CONFIGURE A SMALL OFFICE NETWORK

Assignment - 14



1. **Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.**

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I created a network topology in Cisco Packet Tracer using one switch, three desktop PCs, and two laptops. All five devices were connected to the same switch and assigned static IP addresses within the same subnet (192.168.1.0/24).

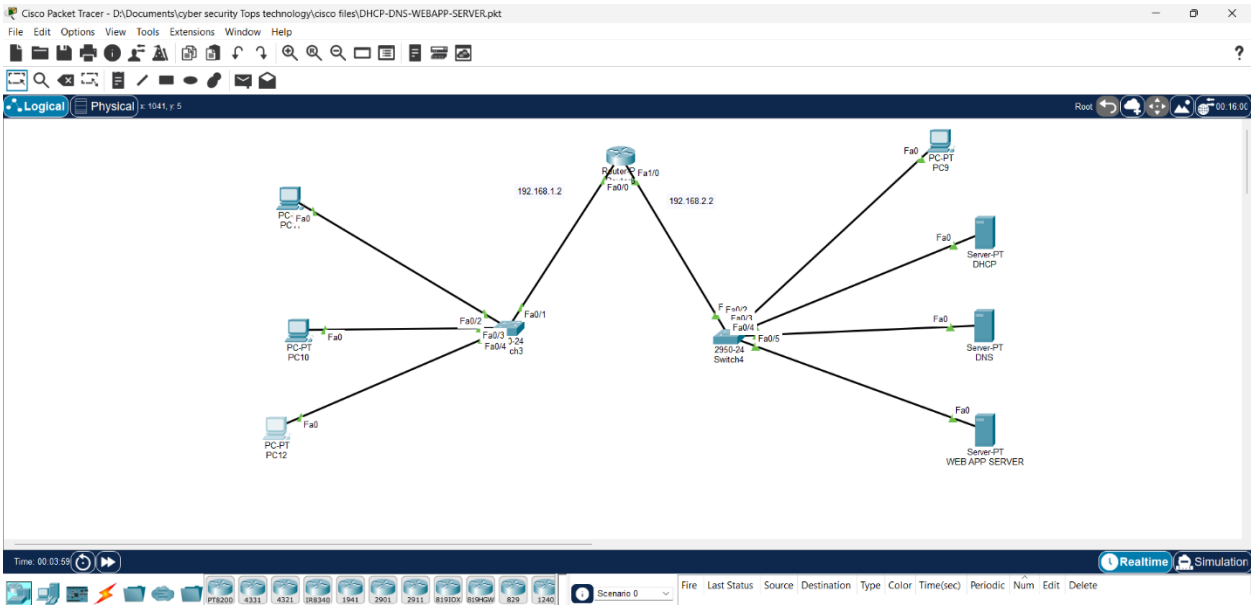
The IP addresses assigned were:

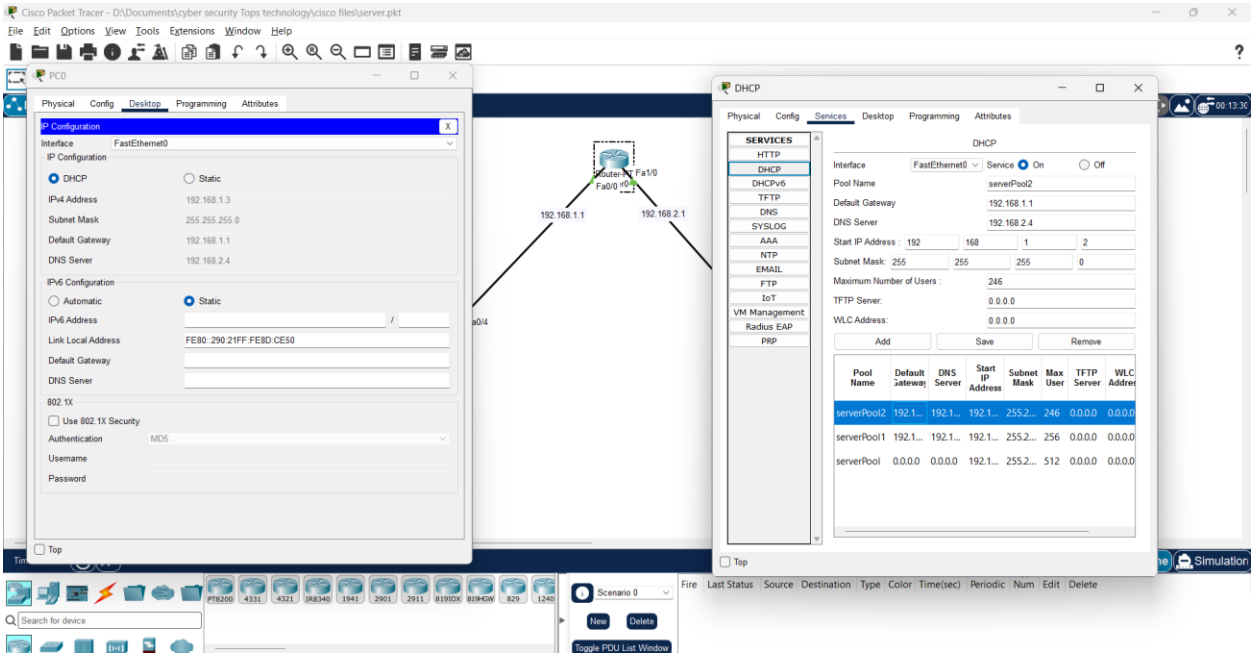
- **PC0: 192.168.1.1**
- **PC1: 192.168.1.2**
- **PC2: 192.168.1.3**
- **Laptop0: 192.168.1.4**
- **Laptop1: 192.168.1.5**

To verify connectivity, I used the ping 192.168.1.2 command from Laptop1's command prompt. The test was successful, with 4 packets sent, 4 packets received, and 0% packet loss.

2. **Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.**

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DHCP Server Configuration and IP Address Assignment

I configured a DHCP server in Cisco Packet Tracer and connected multiple devices through network switches and a router. The DHCP service was enabled, and a DHCP pool was created with a specific IP address range, subnet mask, default gateway, and DNS server.

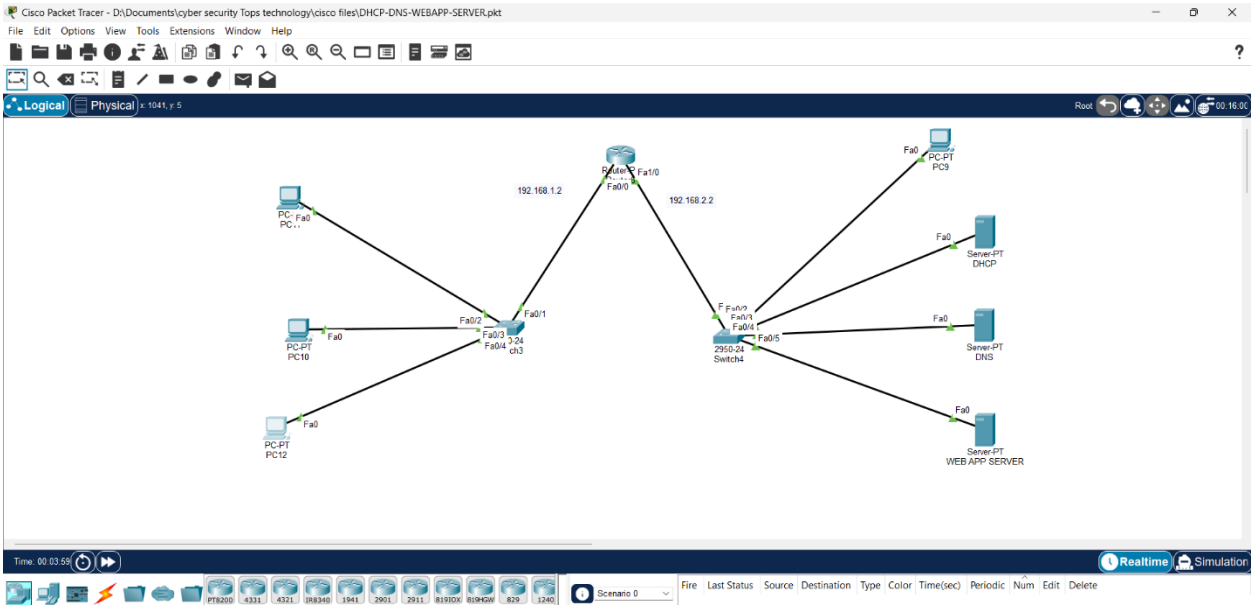
The connected devices were configured to obtain IP addresses automatically using DHCP instead of assigning them manually.

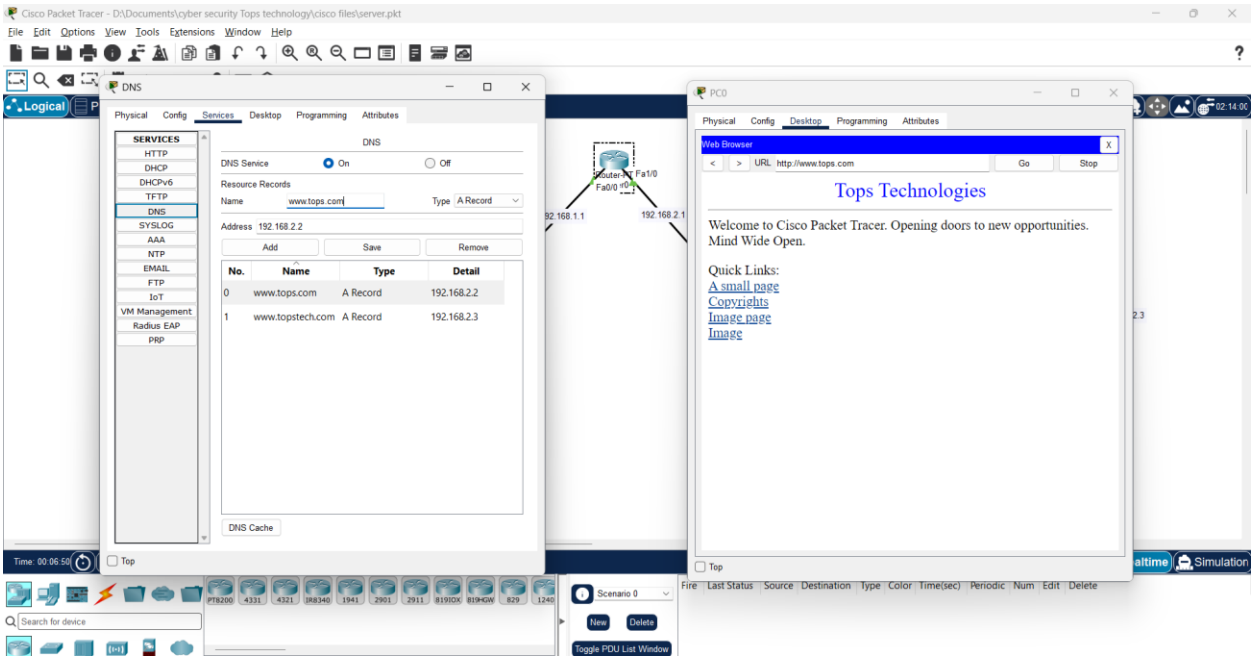
The DHCP server automatically provides IP addresses to the connected devices from the configured address pool. Each device receives a unique IP address within the correct subnet.

Result: The DHCP server was successfully configured to assign IP addresses automatically, reducing manual configuration and making network management easier.

3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.
   Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

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DNS Service Configuration and Testing

I configured a DNS server in Cisco Packet Tracer and enabled the DNS service. I created DNS A records to map custom hostnames to their corresponding IP addresses.

The following DNS records were configured:

- **www.tops.com → 192.168.2.2**
- **www.topstech.com → 192.168.2.3**

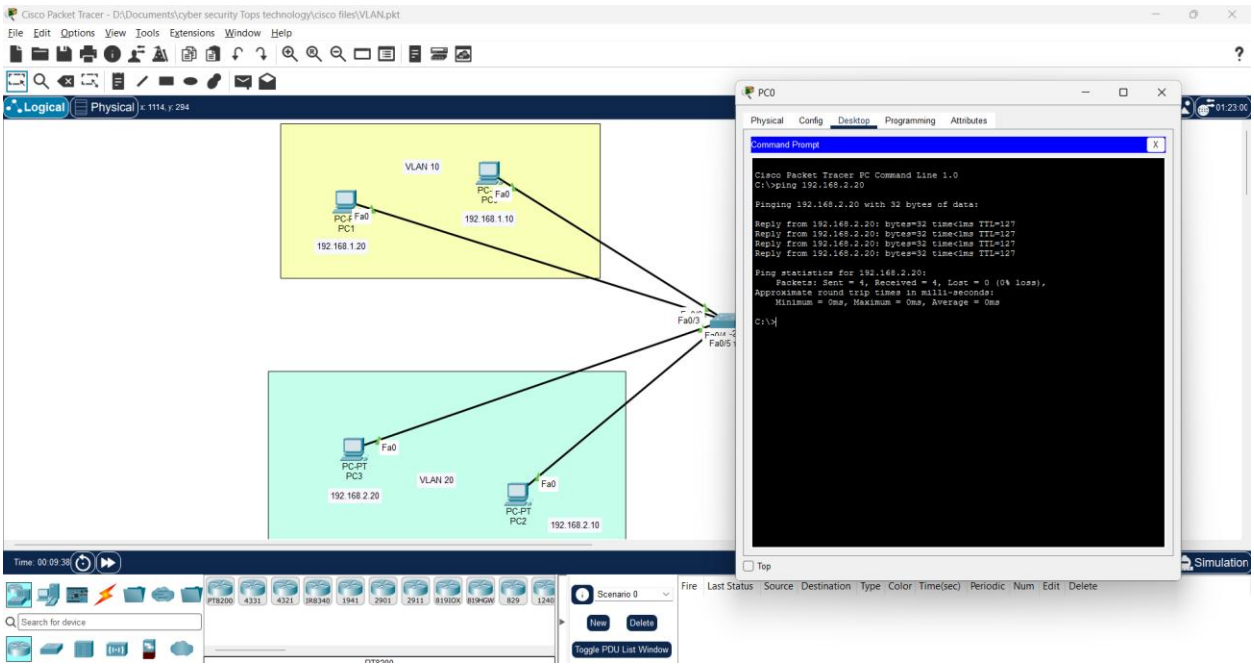
After configuring DNS, I opened the web browser on a connected PC and entered http://www.tops.com. The Tops Technologies webpage opened successfully.

This confirmed that the DNS server correctly resolved the hostname into its corresponding IP address and allowed the client to access the local web page using a domain name.

Conclusion: DNS configuration was successful. It makes network access easier by allowing users to access websites using domain names instead of remembering IP addresses.

4. **Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.**
   Hint: Use ping or file sharing to test connectivity between devices in different VLANs.

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VLAN Configuration and Connectivity Testing

I created two VLANs in Cisco Packet Tracer using a network switch.

VLAN Configuration:

- **VLAN 10 – Workstations**
  - **PC0: 192.168.1.10**
  - **PC1: 192.168.1.20**
- **VLAN 20 – Smart Devices**
  - **PC2: 192.168.2.10**
  - **PC3: 192.168.2.20**

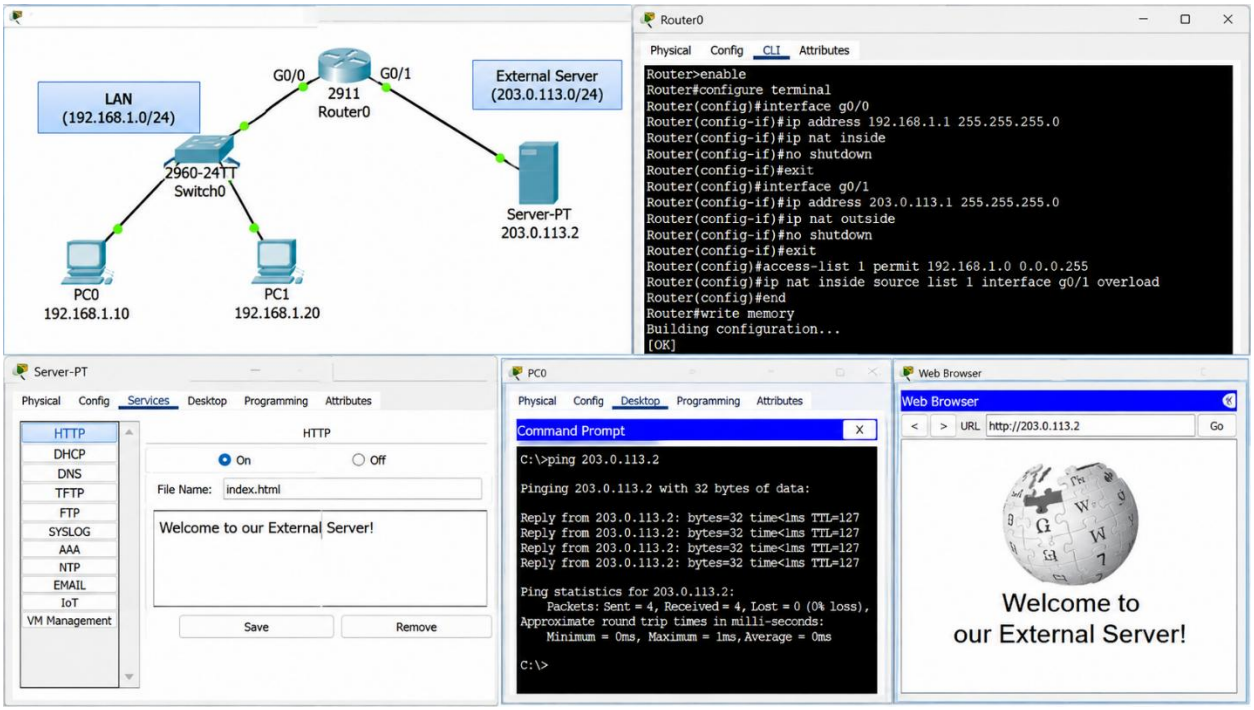
Each VLAN was assigned a separate IP subnet. I used the ping command to test connectivity between the devices.

The devices were able to communicate across VLANs in the current configuration, as shown by the successful ping test.

Conclusion: VLANs help divide a network into separate logical groups, improve network organization, and provide better network management. Communication between different VLANs can be controlled using appropriate routing and access control configurations.

5. **Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.**

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I configured a router with NAT (Network Address Translation) in Cisco Packet Tracer to simulate internet access for a local network.

Two PCs were connected to a switch, which was connected to the router. An external server was connected to the router to simulate an external network.

I configured NAT Overload (PAT) on the router to translate private IP addresses into a public IP address. The external server's HTTP service was enabled to provide a webpage.

To test the configuration, I used the ping command from PC0 to the external server (203.0.113.2). The test was successful, with 4 packets sent, 4 received, and 0% packet loss. I also accessed the external server's webpage using the web browser.